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JELLE DE BOER

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JELLE DE BOER *Department of Geology, Wesleyan University, Middletown, Connecticut*

# Paleomagnetic Differentiation and Correlation of the Late Triassic Volcanic Rocks in the Central Appalachians (with Special Reference to the Connecticut Valley)

**Abstract:** During the late Triassic and early Jurassic, the location of the earth's magnetic pole changed rapidly. As a result of these changes the inclination of the magnetic field in North America increased considerably in a relatively short time. The igneous rocks emplaced during the late Triassic and early Jurassic, show thermo-remanent directions (T.R.M.) of magnetization with progressively higher positive inclinations.

In Connecticut four periods of late Triassic-early Jurassic volcanic activity have been distinguished geologically. They are from old to young: the Talcott, Holyoke, Hampden, and Higganum volcanic events. The mean inclination of the T.R.M. is + 12° for the Talcott, + 25° for the Holyoke, + 42° for the Hampden, and + 35° for the Higganum volcanic units.

This paleomagnetic stratigraphy enables one to differentiate between the volcanic units and to correlate the units of one basin, with contemporaneous outflows or intrusions in other basins.

By means of their distinctive paleomagnetic data it is possible to correlate the volcanics of the Newark and Gettysburg basins (Pennsylvania) with the Holyoke volcanic event, and the volcanics of the Bay of Fundy (Nova Scotia) with the Hampden volcanic event of Connecticut. Such correlations suggest the possibility of a northeastward displacement of the volcanic activity in late Triassic time. This would indicate a progressive northeastward expansion of the broad geanticlinal arching of the Appalachians in the early Mesozoic time.

The origin of the rapid shift is unknown. It appears, however, that the shift is related to a high order reversal of the magnetic field. The shift occurred in an interval of relatively low field strength separating the Permian positive magnetic era from the Cretaceous negative magnetic era. The same interval was characterized by an increase in the frequency of lower order reversals (magnetic epochs and events).

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## INTRODUCTION

In the highly faulted Triassic of central Connecticut certain isolated exposures of basaltic rock cannot easily be ascribed to one of the three principal lava-flow units or to one of the many smaller intrusives. These uncertainties became problematic during the recent increase in detailed geological mapping. For stratigraphic as well as structural purposes, it is necessary to be able to differentiate among the various volcanic rocks. Because the composition of the rock is so uniformly basaltic, reliable petrographic identification has not proved possible. Field geologists with some experience could recognize a flow unit by specific textural and structural characteristics. Small outcrops and samples from drill holes, however, will seldom yield enough information concerning these larger-scale features.

More detailed knowledge concerning the sequence of volcanic events in the different Triassic basins of the Appalachians is desirable. To establish such a volcanic history, a relatively quick and cheap method of correlation is needed. The absolute dating method is not accurate enough to provide the time-resolution for the close age range of the lava flows. Because of the special magnetic conditions in the late Triassic and early Jurassic, paleomagnetism can be used for detailed relative dating, differentiation, and correlation of the igneous rocks.

## ACKNOWLEDGMENTS

I wish to thank Dr. D. E. Bowker for permitting me to use the extensive paleomagnetic data that he submitted to the Massachusetts Institute of Technology for a Ph.D. thesis. I am grateful to Mr. H. W. Krueger of Geochron Laboratories Inc., who permitted the use of several unpublished isotopic data obtained from lava flows in the Connecticut graben. Peter Shive, Roger Young, and David King will contributed to this paper by participating in an undergraduate research project, financed by the National Science Foundation.

## GEOLOGICAL SUMMARY FOR THE CONNECTICUT BASIN

The Triassic basin of southern New England, also known as the Connecticut Valley (or rift zone), is a downfaulted crustal segment with dimensions of 5 to 20 miles by 105 miles (Fig. 1). This north-south-trending segment is located between the eastern and western Highlands, which are chiefly composed of early Paleozoic (and possibly older) metamorphic rocks. The Connecticut Valley consists structurally of two basins, each with its own sedimentary and volcanic record. The northern or Deerfield basin (northern Massachusetts) is the smaller of the two and it can be best considered a northern splay of the Hartford basin (southern Massachusetts and Connecticut). West of Amherst the two basins are separated by the Mount Warner inlier of metamorphic rocks.

The first investigations of the Triassic rocks in the Connecticut Valley were carried out in the middle of the nineteenth century. Percival's report (1842) is the most outstanding of these studies. Even today it is widely respected for its completeness and detail. The foundation for the present-day theories on the origin of the Triassic rocks was laid down by Davis (1898). Davis believed that the Triassic sediments were deposited in a synclinal basin formed by downwarping. The gradual subsidence kept pace with a slow upheaval of the Highlands. The detritus was introduced into the basin from the adjacent crystalline Highlands.

Dana's publication (1891) reflects the opinion in those days that most igneous rocks found intercalated in the Triassic deposits were of intrusive origin. Davis was able to prove that three different lava-flow units existed, which could be distinguished from intrusive sills. Davis's ideas were only slightly modified in later years. The most significant of these modifications was made by Krynine (1950), who proved that most sediments were derived from the eastern Highlands and that they were deposited as a series of coalescing alluvial fans by west-flowing streams.

The upper Triassic or Newark Group of the Hartford basin is subdivided into three formations: the New Haven Formation, the Meriden Formation, and the Portland Formation. The Meriden Formation consists of a succession of

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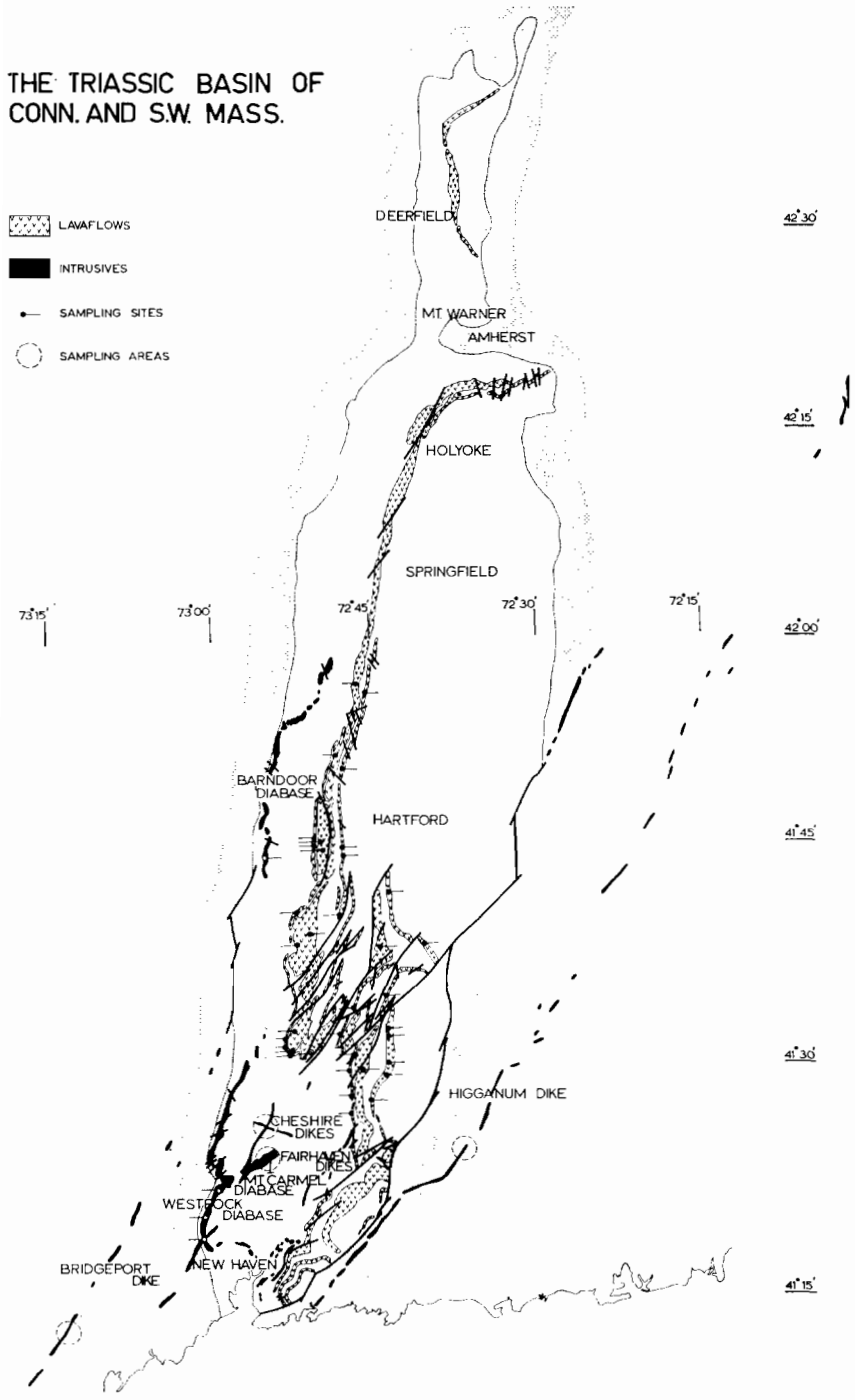



Figure 1. The Triassic basin of Connecticut and SW-Massachusetts. Indicated are the sampling sites studied by Young, Shive, Kingwill, and de Boer.

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three volcanic units separated by sediments. In chronological order these are: the Talcott volcanic unit, the Shuttle Meadow sedimentary unit, the Holyoke volcanic unit, the East Berlin sedimentary unit, and the Hampden volcanic unit.

The Newark Group of the Deerfield basin has also been subdivided into three formations: the Sugarloaf Formation, the Deerfield Formation, and the Mount Toby-Turners Falls Formation. The Sugarloaf and Mount Toby-Turners Falls Formations are generally considered to be the stratigraphic equivalents of the New Haven and Portland Formations of the Hartford basin. The Deerfield volcanic unit has been stratigraphically correlated with the middle and most massive of the three volcanic units in the Hartford basin, the Holyoke unit.

In addition to the volcanic units mentioned above, the volcanic activity of the Connecticut Valley included injection of several types of intrusives. The most obvious of these intrusives are the slightly curved subvertical (giant dikes) and subhorizontal (sills) sheets of basalt in the southern part of the Hartford basin. Also several normal-sized dikes crop out in the southern part of this basin and in the adjacent crystalline Highlands. Because of their petrographic similarities it has not been possible so far to correlate intrusives and lava flows.

The structural history of the Connecticut Valley has been worked out by Sanders (1963), who distinguished four late Triassic-early Jurassic tectogenetic phases: (1) initial period of graben subsidence; contemporaneous sedimentation and basaltic igneous activity (emplacement of flows, giant dikes, and sills); (2) longitudinal arching of the central Appalachians, causing a tilting of the sedimentary deposits and igneous units; (3) second period of graben subsidence; no sedimentation nor volcanic activity; development of transverse folds; and (4) final phase of faulting, followed by injections of basaltic magma, resulting in normal sized dikes; development of longitudinal folds.

Of specific interest for paleomagnetic purpose is the proposed existence of two periods of basaltic igneous activity, separated by a period of deformation.

## ISOTOPIC DATING OF THE TRIASSIC IGNEOUS ROCKS

The absolute dating of basalts is still suffering from the uncertainties concerning the retentivity of Ar and He. The relative abundance of common Sr and the poverty of K in the basalt further set back the different dating possibilities. The low retentivity of He has more or less excluded the use of the U/Pb method, and the high common Sr content that of the Rb/Sr method for basalts.

A study of the Palisades basalt by Erickson and Kulp (1961) indicated that proper sampling may make it possible to use the K/Ar method. In the diabase, retentivity differences up to 25 percent were found. These variations appeared to be related to texture. The retentivity was 100 percent for samples taken from the contact zone of the basalt and from the hornfels, and 75 percent for the coarse inner part of the intrusion. The relationship between retentivity and texture did not apply to whole rock samples taken from the Watchung flows (New Jersey). The retentivity in these rocks was as low as 40 percent. Krueger of Geochron Laboratories, Inc.,[1] is of the opinion that no significant Ar loss occurred during crystallization. He found a retentivity of close to 100 percent for whole rock analyses of samples taken from the Holyoke and Deerfield flows in southwest Massachusetts. Krueger kindly permitted the author to include here the age determinations obtained for the latter flows.

The following reliable dates have been established so far for the igneous rocks of the Triassic basins in the Appalachians:

| | |
|---|---|
| *Erickson and Kulp* (1961) | |
| Palisades intrusion, biotite | 190 ± 5 m.y. |
| Palisades intrusion, whole rock at contact | 202 ± 10 m.y. |
| Palisades intrusion, hornfels at lower contact | 193 m.y. |
| *Krueger Geochron Lab., Inc.* (unpubl.) | |
| Holyoke lava-flow unit, near type locality | 193 ± 6 m.y. |
| Deerfield lava-flow unit, near type locality | 191 ± 6 m.y. |

The age differences between these absolute ages are relatively small. Unfortunately no dates are known for the Talcott and the Hampden flow units. It would be of great interest if some dates could be obtained from the contact zones of these flow units, so that the period of late Triassic volcanism could be established more accurately.

The paleomagnetic results obtained by Opdyke and Wensink (1966) for the plutonic and

[1] Mr. Krueger of Geochron Laboratories Inc., in Boston, permitted the author to use his unpublished radiometric dates of the Holyoke and Deerfield lava flows in Massachusetts.

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volcanic rocks of the White Mountains of New Hampshire are mentioned in this study. Hurley and others (1960) obtained concordant K/Ar, Rb/Sr, and U/Pb ages of an average 180 m.y. for the Conway granite of this complex. The paleomagnetic results of these early Jurassic rocks permit us to determine more accurately the rate of polar shift in the late Triassic and early Jurassic.

## PALEOMAGNETIC PROCEDURE

To date, some 500 oriented cores have been collected and measured from sedimentary and igneous rocks in the upper Triassic Newark group of the central Appalachians. The studies of these rocks date back to 1957. The instruments used in the earlier studies were astatic magnetometers. The more recent measurements have been made with spinner magnetometers. The magnetometer used at Wesleyan University is of the spinner type, designed and described by Doell and Cox (1965).

The mineral responsible for the magnetism of most of the igneous rocks is a finely dispersed titanomagnetite in the ground mass. Kobayashi and Schwarz (1966) found a Curie temperature of 545°C for a sample taken from the Hampden flow. From this they concluded that the titanomagnetite has more than 5 mol. percent of titanium.

The initial measurements show conclusively that the samples possess significant amounts of unstable magnetization. Concentrations of magnetic directions that streak toward the present direction of the earth's field are common. To get rid of this viscous magnetization, most samples have been subjected to alternating field demagnetization with peak values ranging from 75 Oe to 500 Oe depending on the standards set by the different authors. A peak value of 150 Oe appears to be sufficient to clean 75 percent of the rocks. Higher values, however, must be used for the dike rocks. Aside from this magnetic correction, a tectonic correction has been applied to the lava flows and sills. Since the monoclinal dip of the Triassic deposits is well known, a simple rotation of maximal 30° around the strike line is usually sufficient. The different trends of the late Triassic basins are due to faulting along preexisting sinuosities in the metamorphic basement and have not been caused by post-Triassic rotations of crustal segments. Therefore, it was not necessary to make a correction for possible rotations along subvertical axes.

## PALEOMAGNETIC DATA

To answer criticism concerning the reliability of paleomagnetic data, Du Bois and others (1957) made a combined study of some late Triassic rocks in the United States. This study yielded the first paleomagnetic data for the late Triassic rocks in the Connecticut graben. The most significant result of the study was that paleomagnetic data from both volcanic and sedimentary rocks, deposited or emplaced at approximately the same geological time at localities some thousands of miles apart within the same continent, are in agreement. More detailed papers, resulting from these preliminary measurements, were published in 1961 by Irving and Banks for the Massachusetts Triassic and by Opdyke for the New Jersey Triassic. In 1960 Bowker finished an extensive paleomagnetic study of the Triassic rocks throughout the Appalachians, which was used for a Ph.D. thesis at M.I.T. His results yield important data.

The initial studies of Du Bois and others (1957) yielded a Triassic magnetic pole for the Connecticut graben, located at 55°N. 88°E. This pole position was confirmed by Irving and Banks. Opdyke (1961), however, obtained a Triassic magnetic pole for the New Jersey Triassic located at 63°N., 108°E., and Beck (1965) found a Triassic magnetic pole for the Pennsylvania Triassic situated at 62°N., 105°E. The differences between these two sets of Triassic magnetic poles are about 10° in latitude and approximately 15° in longitude. Bowker (1960) obtained eight magnetic poles for the Triassic igneous rocks in Massachusetts and Connecticut. The different pole positions vary from lat 43°N. to 75°N. and long 77°E. to 30°W. This large variation is most probably caused by incomplete magnetic cleaning of the rocks.

Irving and Banks (1961) found that the inclinations of the N.R.M. (natural remanent magnetization) directions measured in rocks sampled from the Holyoke and Granby volcanic units (in Massachusetts) varied from $-10°$ to $+82°$. This variation led these authors to the conclusion that most of the rocks had a V.R.M. (viscous remanent magnetization) which was imposed recently and consequently had a direction parallel to that of the present dipole field in Connecticut. A treatment of the specimen in an alternating current magnetic field with a peak value of 300 Oe reduced the scatter considerably, but still left inclinations varying from $+1°$ to $+37°$.

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Measurements of the N.R.M. in the Triassic igneous rocks of Connecticut by the author and students yielded magnetic directions with inclinations varying from +10° to +70°. Initially it was thought that the samples, which had been subjected to demagnetization fields with a peak value of 150 Oe, were insufficiently demagnetized. The magnetic directions with the high inclinations lie close to the direction of the present dipole field in Connecticut. One would expect the magnetic vector to rotate away and acquire lower inclinations during the demagnetization. The rocks were, therefore, subjected to higher fields (up to 700 Oe) and though the scatter of the magnetic directions did reduce, the ultimate direction of the stable and presumably pure T.R.M. still had inclinations up to +55°.

It appears, therefore, that the large scatter in magnetic directions is not due to a V.R.M. that is hard to remove, but is rather a distinct property of the T.R.M. of the Triassic igneous rocks. This result looked disappointing. So far, however, the late Triassic data had always been plotted together, resulting in a large concentration of data which could hardly serve to obtain a reliable magnetic pole. A surprise came when the magnetic data for the different volcanic flow units and intrusions were plotted separately. It was found that the different volcanic flow units exhibited a distinct range in values for the inclination of the T.R.M. (Fig. 2).

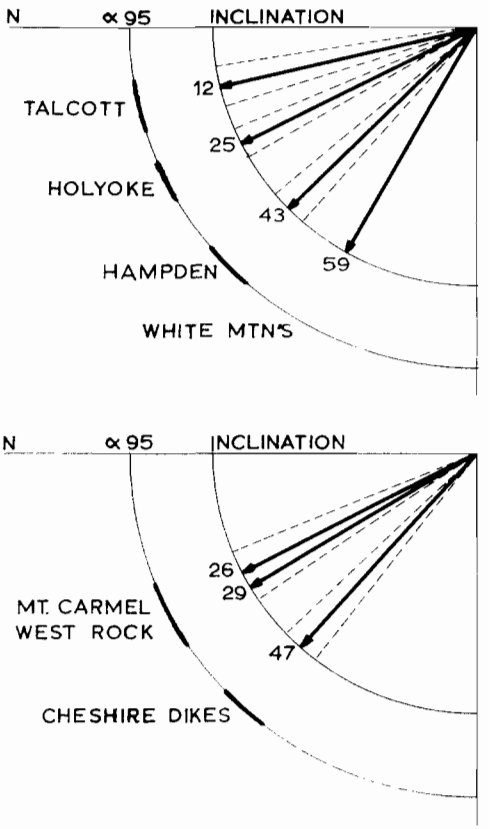



Figure 2. Variations in the inclination of the paleomagnetic directions obtained from the late Triassic and early Jurassic volcanic rocks in Connecticut and New Hampshire.

*Differentiation of the Triassic Igneous Rocks in the Connecticut Valley*

The following values have been measured for the inclination of the T.R.M. in samples collected from the three different lava-flow units which are intercalated in the late Triassic sediments of the Meriden Formation (middle Newark Group).

THE TALCOTT VOLCANIC UNIT: 10 outcrops, 51 cores, 111 samples; inclination variation +7° to +20° (Fig. 3A); average inclination +12° (Table 1).

THE HOLYOKE VOLCANIC UNIT: 10 outcrops, 81 cores, 150 samples; inclination variation +15° to +34° (Fig. 3A); mean inclination +25° (Table 1).

THE HAMPDEN VOLCANIC UNIT: 14 outcrops, 76 cores, 146 samples; inclination variation +31° to +51° (Fig. 3A); mean inclination +42° (Table 1).

Geological evidence points to a possible existence of several flows within one volcanic unit. In central and northern Connecticut, however, it is difficult to distinguish these flows.

The complex Talcott unit (100 to 400 ft thick) consists of probably two lava flows in the central part of the Hartford basin. In the southern part Sanders (1963) was able to distinguish four distinctive basalt flows, separated by sediments. Outcrops of the latter flows, however, are either very brecciated or weathered. Therefore the cores were collected in the central and northern part of the Hartford basin only.

The Holyoke volcanic unit is the thickest (300 to 600 ft) of the three volcanic units involved. It consists of probably two, possibly more lava flows. Because the flows are not separated by sedimentary sequences, they are hard to distinguish. The youngest or Hampden volcanic unit (50 to 150 ft thick) is quite massive

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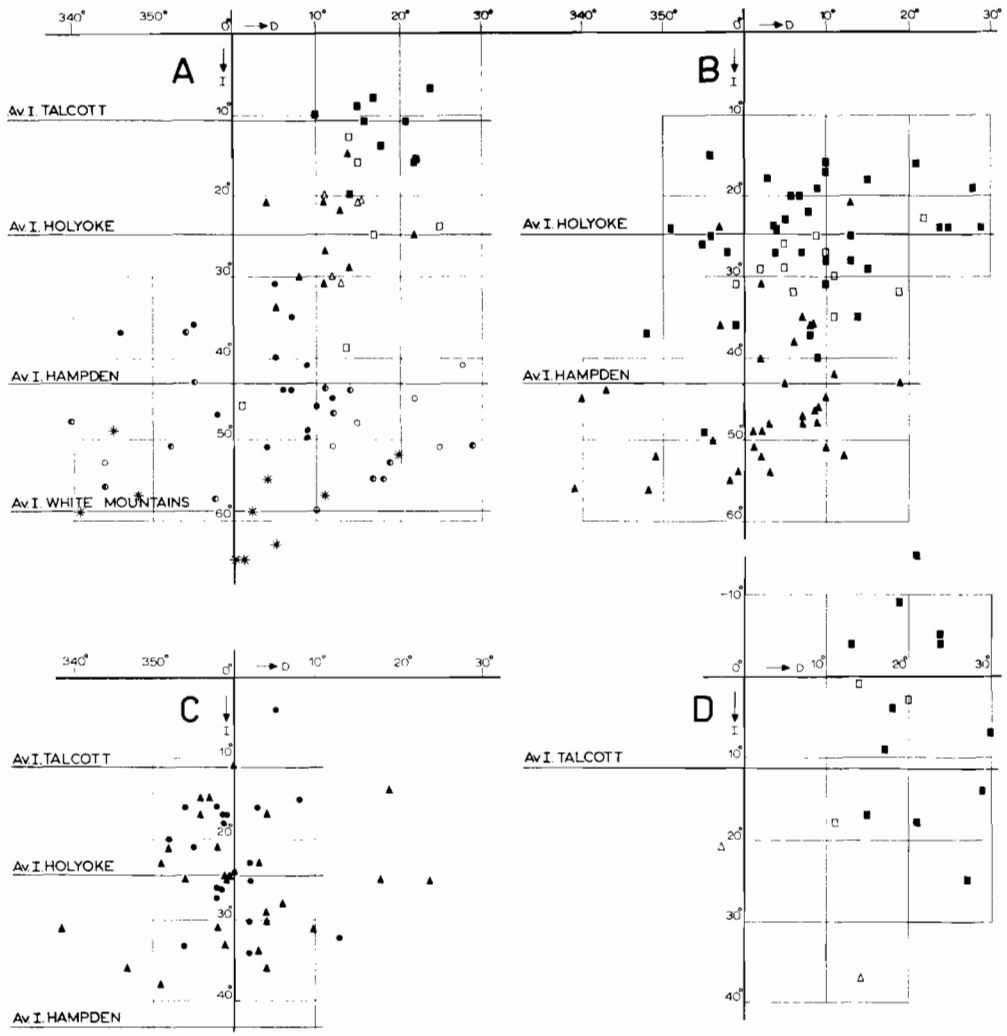



Figure 3. Site means of the paleomagnetic directions obtained from late Triassic and early Jurassic volcanic rocks in the central Appalachians.

(A) open square: Talcott volcanic unit (Bowker)
solid square: Talcott (de Boer)
open triangle: Holyoke volcanic unit (Bowker)
solid triangle: Holyoke (de Boer)
open circle: Hampden volcanic unit (Bowker)
solid circle: Hampden (de Boer)
half solid circle: Hampden (Young)
asterisk: White Mountain volcanics (Opdyke and Wensink)

(B) open square: West Rock intrusives (de Boer)
solid square: Mount Carmel intrusives (Kingwill)
solid triangle: Cheshire dikes (Shive)

(C) solid circle: Volcanics of Gettysburg basin (Beck)
solid triangle: Volcanics and sediments of Newark basin (Opdyke)

(D) solid square: Deerfield volcanic unit (Bowker)
open square: Holyoke volcanic unit (Irving and Banks)
open triangle: Granby volcanic unit (Irving and Banks)

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TABLE 1. PALEOMAGNETIC DATA FOR THE LATE TRIASSIC VOLCANIC ROCKS OF THE CENTRAL APPALACHIANS

| | | C/S | D | I | $\alpha_{95}$ | k | pole Lat N | pole Long E |
|---|---|---|---|---|---|---|---|---|
| 1957 Du Bois and others | Triassic Conn. | 12/— | 12 | +14 | 15° | .. | 55 | 88 |
| | Triassic Mass. | 8/— | 10 | +14 | .. | .. | 54 | 90 |
| 1960 Bowker | Deerfield flow Mass. | 8/16 | 19 | −1* | 5° | 105 | 44 | 80 |
| | Deerfield flow Mass. | 8/13 | 34 | +29* | 11° | 24 | 50 | 50 |
| | Hampden flows Conn. | 8/12 | 13 | +47* | 6° | 72 | 72 | 69 |
| | Talcott flows Conn. | 6/7 | 15 | +27* | 9° | 57 | 58 | 80 |
| | Talcott flows Conn. | 6/8 | 4 | +75* | 15° | 22 | 70 | 67W |
| | Holyoke flows Conn. | 3/5 | 15 | +19* | 5° | 625 | 63 | 75 |
| | North Mountain flows Nova Scotia | 73/216 | .. | . | 10° | 20 | 78 | 77 |
| 1961 Opdyke | Intrusions N. J. | 17/— | 356 | +28 | .. | 38 | 63 | 108 |
| | Watchung flows N.J. | 13/— | 9 | +23 | .. | 100 | | |
| | Sediments N.J. | 48/— | 357 | +23 | .. | 46 | | |
| 1961 Irving and Banks | Holyoke flows Mass. | 10/20 | 10 | +16 | .. | .. | 55 | 88 |
| | Granby tuffs Mass. | 6/12 | | | | | | |
| 1965 Beck | Birdsboro Intr. Penn. | 33/— | 359 | +23 | 3° | 105 | 62 | 101 |
| | Gettysburg Intr. Penn. | 23/— | | | | | | |
| | Yorkhaven Intr. Penn. | 39/— | | | | | | |
| 1965 Opdyke and Wensink | White Mtn. Intrusives | 154/— | 358 | ±59 | .. | 110 | 85 | 124 |
| 1966 de Boer [1968] | Talcott Conn. | 51/111 | 18 | +12 | 4° | 185 | 51 | 78 |
| | Holyoke Conn. | 81/150 | 12 | +25 | 3° | 146 | 60 | 83 |
| | Hampden Conn. | 76/146 | 4 | +43 | 4° | 102 | 73 | 92 |
| | West Rock Conn. | 18/40 | 9 | +29 | 4° | 136 | 63 | 89 |
| Young | Hampden Conn. | 13/54 | 0 | +49 | 7° | 34 | 79 | 106 |
| Kingwill | Mt. Carmel Conn. | 33/67 | 10 | +26 | 4° | 32 | 61 | 87 |
| Shive | Cheshire dikes Conn. | 41/41 | 1 | +47 | 4° | 31 | 76 | 103 |

* These data were recalculated with the permission of Dr. Bowker. Instead of introducing one vector for each sample, a mean vector has been determined for each core.

*Key to symbols:*

C/S — number of cores and samples derived from these cores
D I — declination and inclination of mean magnetic vector
k — precision parameter
$\alpha_{95}$ — radius of the circle of confidence at the 95 percent probability
pole — paleomagnetic pole position

in the northern and central part of the Hartford basin. In the southern part of the basin the unit consists of two flows separated by about 20 ft of siltstone. A detailed study of the apparently massive unit in the central and northern part of the basin by Chapman (1965) revealed the presence of eight sheets (flows?), with distinctive textural and structural differences.

If different flows do exist within one volcanic unit, one would expect changes in the direction of T.R.M. from one flow to the other, because of the possible influence of the secular variation of the Triassic magnetic field. To check this possibility a detailed study was made of the Hampden volcanic unit in the road cut of Route 72 near Berlin. Chapman (1965) distinguished eight sheets in this apparently massive unit, which were correlatable over some 20 miles in a north-south section. The thickness of the sheets ranges from 4 to 44 ft. The top and bottom parts of the sheets are finer grained than their centers, and the sheets are occasionally separated by a thin layer of brecciated basalt. The upper contacts are usually marked by continuous bands of vesicles. The fresh rock in the contact zone is oxidized to red or purple. These observations favor the assumption that the sheets represent different flows.

From each sheet, nine samples were collected for paleomagnetic purpose, three from the bottom part, three from the center, and three from the top part. The paleomagnetic measurements showed that no significant differences existed

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between the T.R.M.s of the different sheets (see Table 2). The variation in declination from sheet to sheet is maximal 15°, the variation in inclination maximal 14°.

The author is of the opinion that the above-mentioned variations are due not to secular variations of the Triassic magnetic field, but to inaccuracies in the measurements and correction.

This means that the time interval, during which the sheets in the Hampden volcanic unit were emplaced, must have been relatively short. The latter conclusion warrants the possibility that the sheets, with their distinct textural and structural differences, originated by magmatic segregation during laminar flow and that they do not represent different lava flows.

To evade the possibility of directional differences within one volcanic unit, the flow(s) was sampled in each outcrop over its total thickness. With a few exceptions the magnetic directions in such sections showed variations which were not significantly different from those found in the Hampden basalt of Route 72. The Hampden volcanic unit in an outcrop, west of Branford where two basalt flows are separated by a few feet of maroon siltstone, shows a similar relatively small variation in directions. It has generally been assumed that

TABLE 2. COLUMNAR SECTION OF HAMPDEN BASALT AT BERLIN, CONNECTICUT

| | thickness | C/S | D | I | $\alpha_{95}$ |
|---|---|---|---|---|---|
| Sheet 8 | 4 + m. | 9/12 | 355 | +45 | 8 |
| Sheet 7 | 3.5 | 9/10 | 002 | +37 | 5 |
| Sheet 6 | 3 | 9/14 | 010 | +38 | 7 |
| Sheet 5 | 2 | 9/11 | 359 | +34 | 4 |
| Sheet 4 | 2.5 | 9/9 | 009 | +48 | 4 |
| Sheet 3 | 1 | 9/16 | 356 | +47 | 6 |
| Sheet 2 | 2.5 | 9/19 | 000 | +43 | 3 |
| Sheet 1 | 15 | 9/18 | 007 | +35 | 10 |
| Whole unit | | 72/109 | 002 | +41 | 5 |

the nondipolar part of the magnetic field did change during the Triassic time as it does at present. However, our data suggest that this may not have been the case. In a publication like this one, it would be desirable to use a statistical analysis which shows that the paleomagnetic characteristics of the separate flows within one volcanic unit are not different, but that the volcanic units have significantly different directions.

Such an analysis was not possible for this study, in view of the following: (1) in most outcrops it is impossible to distinguish the different sheets, considered to be separate lava flow, (2) the origin of these sheets is debatable, especially in the light of their magnetic characteristics.

The differences in the mean directions obtained for the different volcanic units are respectively 14° ± 7° (Talcott-Holyoke) and 20° ± 7° (Holyoke-Hampden). These differences are obvious enough to warrant their use for stratigraphic purpose.

In addition to the three flow units, several intrusives were studied which vary widely in shape and size. The most obvious are the supposedly subvertical sheets of the Westrock and Barndoor diabase, which follow the western boundary of the Hartford basin over most of its length. Similar, but subhorizontal sheets form the Mount Carmel and the Rabbit Rock (Foxon area). The Newark Formation has also been intruded by widely spaced swarms of normal-sized dikes such as the Cheshire dikes. The results of the study are as follows.

MOUNT CARMEL SILL: 5 outcrops, 33 cores, 67 samples; inclination variation +15° to +40° (Fig. 3B); average inclination +26° (Table 1); data obtained by Kingwill[2].

WEST ROCK GIANT DIKE: 3 outcrops, 18 cores, 40 samples; inclination variation +23° to +35° (Fig. 3B); average inclination +29° (Table 1).

CHESHIRE DIKES: 41 cores, 41 samples; inclination variation +21° to +72° (Fig. 3B); average inclination +47° (Table 1); data obtained by Shive[2].

FOXON-FAIRHAVEN, BRIDGEPORT, AND HIGGANUM DIKE SYSTEMS: inclination variation +34° to +48°; average inclination +38°.

The Mount Carmel sill is formed by a large saucer-shaped intrusion of basalt, more or less parallel to the east-dipping Triassic strata. It was emplaced in the New Haven Formation, the oldest arkose in the Hartford basin. The West Rock giant dike is a large intrusive in the New Haven Formation. It forms part of a north-trending complex which appears to parallel the western boundary of the Hartford basin. Its structural relationship to this boundary is unknown.

The Cheshire dike system consists of a set of one north-northeast- and one west-northwest-trending dike, which intruded the New Haven arkose. On his geological map of the Mount Carmel quadrangle, Fritts (1963) suggests that

[2] Kingwill and Shive studied the paleomagnetism of the Mount Carmel sill and the Cheshire dikes, respectively. Their reports are available at Wesleyan University, Middletown, Connecticut.

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TABLE 3. PALEOMAGNETIC CORRELATION OF THE LATE TRIASSIC IGNEOUS ROCK IN THE CENTRAL APPALACHIANS

Effusives

| | *New Jersey* | *Connecticut Massachusetts* | *Massachusetts* | *Nova Scotia* |
|---|---|---|---|---|
| Phase C | .. | Hampden | .. | North Mountain |
| Phase B | Hook Mtn.<br>2nd Watchung<br>1st Watchung | Holyoke | .. | .. |
| Phase A | .. | Talcott | Deerfield | .. |

Intrusives

| | *Pennsylvania* | *New Jersey* | *Connecticut Massachusetts* | *Nova Scotia* |
|---|---|---|---|---|
| Phase C | .. | .. | Cheshire d. | Shelburne d. |
| Phase B | Birdsboro s.<br>Gettysburg s.<br>York Haven s. | Sourland Mtn. s.<br>Palisades s.<br>Belle Mtn. s.<br>Baldgate Mtn. s.<br>Haycock s. | West Rock s.<br>Mt. Carmel s. | .. |
| Phase A | .. | .. | Medford d. | . |

*Key to symbols:*
s = sill
d = dike

the west-northwest dike is related to the West Rock and Mount Carmel intrusions. The north-northeast dike (called Buttress diabase by Fritts) intruded along a fault, which cuts and offsets the west-northwest dike, and the Mount Carmel sill.

The Foxon-Fairhaven dike system consists of a linear arrangement of basaltic dike segments in the New Haven arkose (Fig. 1). These segments were emplaced in a fault, which displaced a folded sequence of rocks belonging to the Meriden and New Haven Formations. This indicates that no relationship exists between the dike system and any of the lava flows (Sanders, 1963).

The Higganum and Bridgeport dike systems consist of a linear arrangement of elongated basaltic dike segments in the metamorphic rocks of the eastern and western Highlands (Fig. 1). The segments are dominantly directed north-northeast; locally this trend changes abruptly and becomes east-northeast. In the latter case the dike system continues *en échelon*. The Foxon-Fairhaven, Bridgeport, and Higganum dikes yield similar paleomagnetic data. The inclinations of their magnetic directions permit a correlation with the Hampden volcanic unit. However, the declinations are definitely different and range from N.10°W. to N.30°W., instead of being generally N.10°E. It is the author's opinion that these dikes are post-Triassic and that they intruded in Jurassic time. Their paleomagnetic characteristics are the subject of a separate study (de Boer, 1967).

The following conclusions can be drawn from the data mentioned above. (1) The Talcott, Holyoke, and Hampden volcanic units can be differentiated paleomagnetically. The differences between the Talcott and Holyoke are less conspicuous than those between the Holyoke and Hampden volcanic units.

(2) The sheet-like intrusives of the West Rock and Mount Carmel basalt in the New Haven Formation occurred simultaneously with the effusion of the most massive of the three volcanic units, the Holyoke lava-flow unit.

(3) The paleomagnetic directions of the north-northeast- and west-northwest-trending dikes near Cheshire are similar. This indicates that they intruded within a short time interval. The dikes can be correlated paleomagnetically with the Hampden volcanic unit. This paleomagnetic correlation contradicts Fritts's (1963) opinion that the west-northwest dike of the set is related to the West Rock and Mount Carmel intrusions (see no. 2).

The paleomagnetism of the volcanic units of the Hartford basin have also been studied by Bowker (1960), Irving and Banks (1961), and Young (unpub.). A comparison of their data with those obtained by the author, reveals the following similarities and differences.

TALCOTT VOLCANIC UNIT: Bowker (1960)

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studied two outcrops in the Talcott. From each outcrop six samples were cored. The variation in inclination is from +13° to +83°. It seems evident that this large variation is due to incomplete cleaning (Fig. 3A).

HOLYOKE VOLCANIC UNIT: Bowker (1960) obtained three samples from one outcrop in the Holyoke. The average inclination is +19°, which is comparable with the author's data. Irving and Banks (1961) sampled from three outcrops in the type locality of this volcanic unit near Holyoke (Massachusetts). The average inclinations of the supposedly T.R.M. in these outcrops are +18°, +3°, and +1° (Fig. 3D). These low values for the Holyoke at its type locality do not agree with those obtained from its supposed equivalent in Connecticut. They are, however, comparable with the range in inclinations of +7° to +20° for the Talcott volcanic unit. This could mean that the Holyoke volcanic unit in its type locality is older than its more southern counterpart. In a discussion in 1891 Davis said, "I am inclined to believe that the anterior sheet (Talcott) in Connecticut thickens northward and becomes the main sheet (Holyoke) north of the state line, while the main sheet of Connecticut thins and becomes a subordinate posterior sheet (Granby?) further northward." Davis based his opinion on the location of the main fish beds with respect to the lava flows.

The possibility of time differences between the outflows of basaltic magma from north to south within one unit has become more significant with Schnabel and Eric's study of the Talcott. Schnabel and Eric (1964) found that the Talcott in the Windsor Locks quadrangle seems to be stratigraphically lower than the Talcott in the Tariffville quadrangle. The difference in stratigraphic position is about 150′ over a distance of 1 mile. A possible explanation for this difference is that the topographically continuous unit consists of segments, which flowed out at different times, within one period of general volcanic activity. If such age differences exist, they may account for the different paleomagnetic directions found within one flow unit. Further research is needed.

HAMPDEN VOLCANIC UNIT: Bowker (1960) and Young[3] measured the paleomagnetic directions in four outcrops of the Hampden. Bowker obtained a range in inclination from +38° to +56°; Young's inclinations vary from +36° to +59° (Fig. 3A). These data agree with the range in inclinations found by the author for the paleomagnetic directions in fifteen other outcrops.

DEERFIELD VOLCANIC UNIT: The Deerfield volcanic unit of the Deerfield basin has been stratigraphically correlated with the Holyoke volcanic unit of the Hartford basin. The inclinations of its remanent magnetization, however, vary from −15° to +17° (Bowker, 1960) (Fig. 3D). These low values do not agree with those obtained for the Holyoke unit (+15° to +34°). It is therefore more probable that the volcanic activity in the Deerfield basin took place earlier, contemporaneously with the Talcott event in the Hartford basin. It is significant to note that both the Deerfield unit, north of the Mount Warner inlier, and the Holyoke unit just south of the inlier, have relatively low inclinations.

From this comparison of the paleomagnetic data in the Connecticut Valley it is evident that more detailed work is needed. Lateral age differences appear to exist within one and the same volcanic unit. Because these are fissure eruptions, such differences can be expected.

### *Correlation of the Triassic Igneous Rocks in the Central Appalachians*

Of specific value for a comparison of the Triassic igneous rocks in the central Appalachians are the data published by Opdyke (1961) and Beck (1965). Before a comparison can be made, the difference in inclination that is caused by the difference in geographic latitude of the regions involved has to be determined. The difference between the Triassic isoclines of Pennsylvania and Connecticut for a magnetic pole located at 62° N., 105° E. (Beck, 1965) is 5°. The difference between the paleoisoclines of New Jersey and Connecticut is also approximately 5°.

NEWARK BASIN (Fig. 4): Opdyke (1961) made a detailed study of the late Triassic magnetism in the different rocks of the Newark graben in New Jersey. The Newark Group of the Newark basin is subdivided into the Stockton, the Lockatong, and the Brunswick formation. Intercalated in the Brunswick formation are three volcanic units, known as the Watchung flows. The Triassic sediments are cut by numerous intrusives, the best known of which is the Palisades sill that is exposed along the southern boundary of the graben. Opdyke studied the Brunswick sediments as well as the Watchung flows and some of the intrusives. The inclinations of the C.R.M. in the sedi-

[3] Young studied the paleomagnetism of the Hampden flow unit. His report is available at Wesleyan University, Middletown, Connecticut.

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ments range from +15° to +38° (average +23°). The T.R.M. in the Watchung flows has inclinations varying from +14° to +34° (average +23°); the T.R.M. in the intrusives (including the Palisades sill) has inclinations that vary from +11° to +30° (average +28°). Since igneous rocks of the same age in Connecticut should have acquired a direction of magnetization with an inclination that was about 5° higher, it is possible to correlate the Watchung flows and New Jersey intrusives with the Holyoke volcanic event (average inclination +25°) and the earlier part of the Hampden volcanic event in Connecticut (Fig. 3C).

GETTYSBURG BASIN (Fig. 4): Beck (1965) studied the Triassic intrusives in the Newark Group of the Gettysburg basin. In this region one can distinguish three large intrusive complexes: the Birdsboro, the York Haven, and the Gettysburg basaltic masses. The range in inclinations obtained from the magnetic directions in these intrusives is from +15° to +34°. (Of the 20 inclinations, 18 fall within this range: one value of 4.5° and one of 42° were excluded.)

The average inclination of the T.R.M. of the above-mentioned intrusives is +23°. Igneous

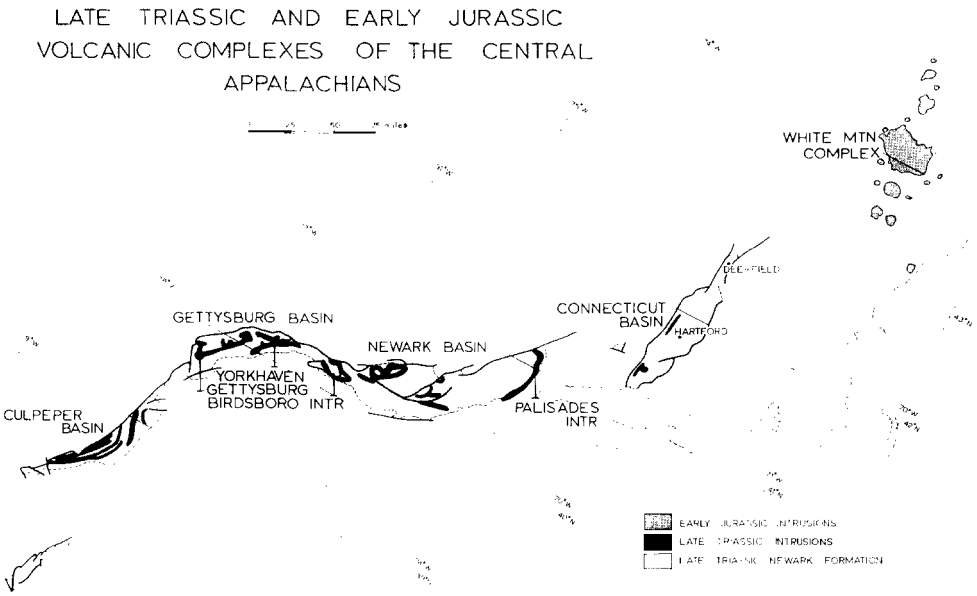



Figure 4.

The average inclination for the three Watchung flows is +23°. A correlation of the first Watchung flow with the Talcott, the second with the Holyoke, and the third with the Hampden is thus not substantiated by the paleomagnetic evidence. The average inclination for the intrusives (including the Palisades) is +28°, which is close to that of the Mount Carmel and West Rock intrusions in Connecticut.

The correlation of the volcanic rocks in the Newark basin with the Holyoke volcanic event in the Connecticut Valley implies that the Triassic volcanism in New Jersey was of relatively shorter duration than that in Connecticut.

rocks of the same age that would have intruded in the Connecticut graben should have acquired magnetic directions with inclinations about 5° larger. The intrusions of the Mount Carmel and West Rock have average inclinations of +26° and +29°, which is an indication that the igneous rocks of southeastern Pennsylvania can be correlated with the Holyoke volcanic event of Connecticut (Fig. 3C). The three large intrusive complexes were accordingly emplaced contemporaneously with the large intrusives in the Newark graben and the Connecticut Valley.

BAY OF FUNDY BASIN: Bowker (1960) col-

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lected a large number of samples from the North Mountain lava flows in Nova Scotia. A correlation of the paleomagnetic directions in Nova Scotia with those in the central Appalachians may be hampered by significant strike-slip movements, which are assumed to have caused a relative northward shift of the peninsula with respect to the mainland in post-Triassic time. These movements, however, could never have caused too large a deviation in the inclination because of the relatively small distance involved. The virtual magnetic pole given by Bowker (see Table 1) is comparable to the ones which were obtained for the Hampden volcanic unit of central Connecticut.

The conclusion that can be drawn from these correlations is that the late Triassic volcanic activity in the central Appalachians was quite close to being a one-phase event. Most of the larger intrusives and the most extensive flows were emplaced during the Holyoke volcanic event (Table 3). In Connecticut the volcanic activity started somewhat earlier (Talcott volcanic event) and lasted somewhat longer (Hampden volcanic event). In the Bay of Fundy basin the volcanism appears to have been later than that in the central Appalachians. This suggests a possible northeastward shift of the centers of volcanic activity in early Mesozoic time. Such a shift may indicate a northeastward expansion of the broad geanticlinal arching of the Appalachians, which resulted in a progressive opening of the fissures, intruded by the mafic magmas.

WHITE MOUNTAIN PLUTONIC-VOLCANIC COMPLEX (Fig. 4): Recently Opdyke and Wensink (1966) published their paleomagnetic study of the White Mountain plutonic and volcanic rocks of New Hampshire. The White Mountain linear series of intrusives and associated extrusives range in composition from gabbroic to granitic. The Conway granite, which is exposed in one of the larger batholiths provided concordant K-Ar, Rb-Sr, and U-Pb isotopic ages of around 180 m.y. (Hurley and others, 1960). Because these granites appear to have intruded in early Jurassic time, a comparison of their paleomagnetic data and those of the latest or Hampden volcanic event in Connecticut may prove useful. The inclinations of the paleomagnetic directions measured in the gabbros, monzonites and mafic dikes of the White Mountain rocks range from 49.5° to 65.5° (Fig. 3A). The average inclination is 59° (Fig. 2). Both negative and positive inclinations were measured. The difference between the early Jurassic isoclines of New Hampshire and Connecticut for a magnetic pole located at 85.5° N., 124.5° E. (Opdyke and Wensink, 1966) is 3°. The inclinations of the magnetic directions in the Hampden volcanic unit range from +31° to +51°. The average inclination is +42°. The paleomagnetic data accordingly suggest that no age relationship exists between the latest volcanic event in Connecticut and the plutonic-volcanic activity in New Hampshire. The New Hampshire results provide important data for the rate of polar shift during late Triassic and early Jurassic time when combined with our results.

*Triassic Reversals of the Magnetic Field*

The late Triassic igneous rocks of the Appalachians were magnetized in a negative or south-seeking field. The inclinations of the paleomagnetic directions are dominantly positive. The low negative inclinations of the magnetic vectors in the Deerfield volcanic unit (Bowker, 1960) can be attributed to either incomplete demagnetization or the proximity of the magnetic equator during effusion. The uniform polarity of the magnetic directions in the upper Triassic rocks of the Newark Formation contrasts sharply with the reversals found by Collinson and Runcorn (1960) in the upper Triassic Chinle Formation of Utah. The results obtained by Picard (1964) in the Chugwater Formation of Wyoming indicate that reversals did occur in early Triassic time. He could distinguish 24 magnetic events within a sequence of 1000 feet of sediments, 12 of which were attributed to changes in the polarity of the earth's magnetic field. From the paleomagnetic results obtained in Europe it is also known that reversals occurred in early Triassic time. The earliest deposits of the south alpine Triassic were magnetized in a reversed field (de Boer, 1963; 1965). This field had a variable polarity almost throughout the Scythian (the Scythian is the interval A of the American time scale). The upper Scythian and younger deposits, however, were invariably magnetized in negative fields.

It seems therefore that though frequent polarity changes were common in the early Triassic, the polarity of the magnetic field remained largely unchanged in late Triassic time.

Kobayashi and Schwarz (1966) recently reported on the measurements of 15 samples collected from two localities in which the contact zone between the Hampden flow unit and East

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Berlin Formation of Connecticut is exposed. One of the conclusions derived is that the sediments of the contact zone were remagnetized while the geomagnetic field was reversed. This should have occurred in the period in which the Hampden acquired its T.R.M. Samples taken from the same localities and from similar stratigraphic positions in other outcrops by the author and by Young have not yielded any indications of reversals. Since both directions given by Kobayashi and Schwarz are far from reverse with respect to the normal Triassic directions (Schwarz implied their polarity from a theoretically projected continuation of the demagnetization path), it is possible that an area with anomalous magnetization was sampled. Such anomalous spots in the basalts have been found several times. They are usually around funnel-shaped accumulations of closely vesiculated (sometimes brecciated) basalt, which most probably resulted by hornito activity.

### *Accelerated Polar Shift During the Triassic in North America*

The late Triassic magnetic poles of the eastern United States form an elongated cluster in a territory now known as central Siberia. The long axis of this cluster points in a north-northeast direction (Fig. 5A). It is highly improbable that this streaking is due to instrumental errors, because the poles were obtained by different authors, using different procedures. It is equally difficult to explain the streaking by incomplete removal of the V.R.M., because most poles were derived from completely and thoroughly demagnetized sequences of rocks. The elongated cluster is therefore thought to be due to a relatively rapid change of the magnetic pole position during the late Triassic and possibly early Jurassic. Opdyke and Wensink (1966) suggested the possible existence of such a phenomenon when they compared the mean inclination of the New Jersey Triassic with the mean inclination of the lower Jurassic White Mountain series. The minimal latitudinal difference between the late Triassic and early Jurassic magnetic poles amounts to ±35°; the longitudinal variation amounts to ±45°. The paleomagnetic poles of the isotopically dated Holyoke flow unit of Connecticut (193 m.y.) and the White Mountain igneous rocks (180 m.y.) are about 25° apart, measured over a great circle. This indicates a polar shift of approximately 10 inches a year, assuming the polar path to have been straight and the movement to have been more or less continuous.

The origin of the relatively rapid shift in pole position is difficult to establish, when the limited period of time is considered. Paleomagnetic data for the late Paleozoic and late Mesozoic, however, may reveal the relation of this shift to earlier magnetic events (Fig. 5B.).

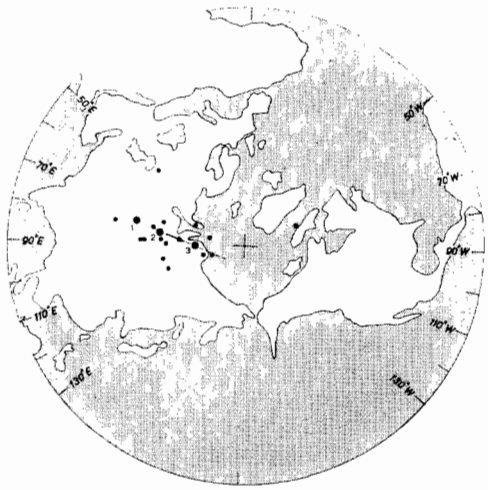



Figure 5A. Virtual magnetic poles for the late Triassic of the central Appalachians. 1, 2, and 3 are the poles, obtained from the Talcott, Holyoke, and Hampden volcanic units of Connecticut.

The Permian was characterized by a positive magnetic field with constant polarity (Fig. 6). The field did not change its polarity for well over 50 m.y. Irving (1966) proposed to call this period the Kiaman magnetic interval (the author prefers the word *era* here; *interval* could then be used for the period between different magnetic eras). From the tight cluster of the virtual magnetic poles for the different continents, it can be concluded that polar movement was minimal in the Kiaman magnetic era. The Kiaman magnetic era accordingly was characterized by a long period of relative magnetic quietness, in which reversal frequency as well as polar wandering were minimal. A very similar magnetic era existed in the Cretaceous. Pole positions determined from the magnetic directions in rocks aged from 138 to 84 m.y. differ only slightly. Grommé and Merrill (1966) concluded that polar wandering relative to North America did not occur in this period. The magnetic field was negative and stayed so

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for well over 50 m.y. It would be useful if Grommé and Merrill proposed a name for this magnetic era, so that it can be used independent from the Cretaceous time interval.

In the 80 m.y. separating the Permian and Cretaceous magnetic eras, reversal frequency increased and polar wandering became a common phenomenon. In 1958 Khramov already




Figure 5B. Virtual magnetic poles for the Permian and Mesozoic of NE-America. A = Permian (Helsley, 1965), B = Triassic (this paper), $C_1$ = Jurassic (Opdyke and Wensink, 1966), $C_2$ = Jurassic (de Boer, 1967), D = Cretaceous (Larochelle, 1961).

foresaw a possible relation between polar wandering and reversal frequency when he concluded that the quicker the co-ordinates of the poles changed, the more often reversals occurred. It is debatable whether the polar wandering was real or apparent and caused by continental drift. The high rates of polar shift during the late Triassic suggest, however, that we are dealing here with polar wandering "sensu stricto." The two phenomena may have intermixed during the Jurassic.

When we consider the above-mentioned data, it appears obvious that, aside from polarity epochs and events, larger order polarity eras can be distinguished, with a periodicity of 50 ± 10 m.y. These magnetic eras are characterized by a constant polarity, little or no changes in field strength and minimal polar shift. Two magnetic eras with opposite polarity appear to be separated by magnetic intervals, which are characterized by frequent field reversals of a lower order (periodicity 2 ± 1 m.y. and 0.1 ± 0.01 m.y.) and by accelerated polar wandering.

The Carboniferous magnetic interval was quite similar to the Triassic-Jurassic interval. The geomagnetic field in Australia had low to moderate inclinations in upper Silurian and Devonian time, and high inclinations from Permian to Mid-Cretaceous (Irving, 1966). Most of the directional changes in this interval occurred in a relatively short period of probably less than 20 m.y., beginning near the end of the lower Kuttung time (Visean of Europe). This magnetic interval with its rapid polar shift resembles very much the late Triassic interval of North America. The Carboniferous interval of Australia marks the transition from the Silurian negative magnetic era to the Permian (or Kiaman) positive magnetic era.

Briden (1966) estimated the variation in field strength during the Carboniferous interval. The dipole field strength, reduced to paleoequator value increased from a low of 0.004 Oe in the upper Devonian, over values of 0.024 and 0.027 Oe to 0.122 Oe in the lower Permian. It appears therefore that variations in field strength occurred contemporaneously with rapid polar shift.

A difficulty arises when the polar paths of Australia and North America are compared. There are no indications for an accelerated polar shift in the Carboniferous of North America, and no indications for a rapid polar shift in the late Triassic of Australia. The European data, however, indicate a Carboniferous as well as a Triassic shift which are comparable to, but smaller than, the North American and Australian shifts. The reason for these differences is most probably that the magnetic imprint of a polar shift depends on the paleomagnetic latitude of the crustal segment involved.

The polar shift is recorded more effectively at lower magnetic latitude than at higher latitude because of the following factors. (1) A shift of the pole over some 20° causes an increase of 35° in the inclination of the magnetic direction in a crustal segment located on the magnetic equator near the plane comprising the polar path. The increase is only 10° near the pole. The latter change is within the limits of error in most paleomagnetic studies.

(2) The change in declination of a magnetic direction is greater near the pole than near the

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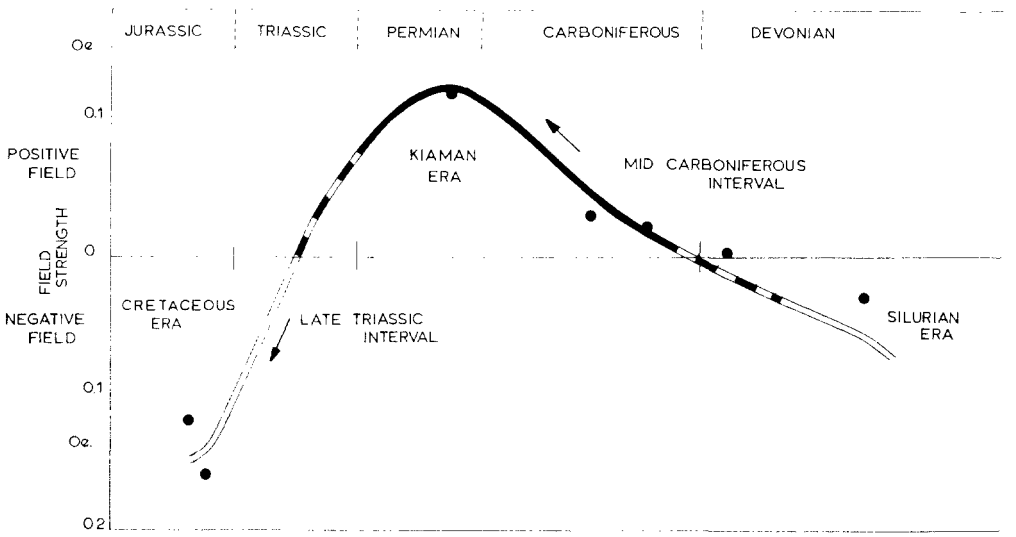



Figure 6. Hypothetical variation in field strength and polarity of the late Paleozoic-early Mesozoic magnetic field (intensity values from Briden, 1966). Periods of frequent lower order polarity changes and accelerated polar shift are indicated by dash line.

equator, for a crustal segment located near the plane normal to the polar path. Because the magnetic declination is determined by the horizontal component of magnetization, it is more difficult to define accurately at high latitudes. The result is that the change in declination caused by the polar shift is difficult to distinguish from the normal deviations in the paleomagnetic directions obtained from rocks magnetized at higher latitudes.

The combined effects of the above-mentioned factors make it difficult to detect a rapid polar shift on the paleomagnetic record of rocks magnetized at high paleolatitude.

The absence of a late Triassic-early Jurassic polar shift on the paleomagnetic record of Australia can be attributed to the relatively high latitude of that continent in the Mesozoic.

The periods of rapid polar shift are of special interest for paleomagnetism because they enable us to build up a refined paleomagnetic stratigraphy, which can be used for detailed differentiation and correlation. If we assume with Simpson (1966) that increased intensity of ionizing radiation during periods of minimal magnetic field strength enhanced the general mutation rate of organisms, then the periods of large-scale animal extinction can be correlated with magnetic intervals and consequently with rapid polar shifts. Paleontological data show approximately six periods of large-scale animal extinction, followed by periods of accelerated evolution, since the Cambrian. The first-mentioned periods mark the boundaries of the Cambrian-Ordovician, the Ordovician-Silurian, the Devonian-Carboniferous, the Permian-Triassic, the Triassic-Jurassic, and the Cretaceous-Tertiary. The low field strength in the upper Devonian (Australia, Briden, 1966) coincides remarkably with the third period of animal extinction. The late Triassic-early Jurassic polar shift (North America and Europe) coincides with the fifth period.

## CONCLUSIONS

The paleomagnetic results of the late Triassic and early Jurassic igneous rocks in the Appalachians indicate that this period was characterized by a relatively rapid shift of the magnetic pole. The result of this shift was that rocks deposited or emplaced in the same region during a short time interval show significantly different paleomagnetic directions. The absolute dating method is not accurate enough to provide the time resolution for the relatively close age range of the lava flows. Because of the significant differences in the magnetic directions, paleomagnetism can be used for relative dating of the volcanic events.

A correlation of the different volcanic units

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in the Triassic rift zones of the Appalachians indicates that the late Triassic volcanism was dominantly a one-phase event. The larger intrusives and the most massive flows were all emplaced during the Holyoke event. By means of its paleomagnetic characteristics the Holyoke event can be separated from the earlier Talcott, and the later Hampden and Higganum events.

The rapid shift of the magnetic pole occurred in an interval of relatively low field strength separating two magnetic eras with opposite polarity. The magnetic eras are characterized by constant polarity, neglectable polar shift, and relatively high field strength. The interval is characterized by an increase in the frequency of lower order reversals (magnetic epochs and events), by accelerated polar shift, and by relatively low field strength. There may be a relation between the magnetic intervals and the paleontological periods of accelerated evolution.

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